Finite Temperature Casimir Effect for Corrugated Plates *

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(Received 5 June 2006)

Using the path-integral method, the corrections to the Casimir energy due to the combined effect of surface roughness and the finite temperature are calculated. For the specific case of two sinusoidally corrugated plates, the lateral Casimir force at finite temperature is obtained. The amplitude of the lateral Casimir force has a maximum at an optimal wavelength of $\lambda \approx 2H$ with the mean plate distance H. This optimal parameter relation is almost independent of temperature.

PACS: 12. 20. Ds, 03. 70. +k, 11. 10. −z

Casimir interactions[1−10] are a fundamental property of vacuum. Since Casimir[1] made the remarkable prediction, great effort has been put into the study of this important subject. Two important extensive discussions are the surface roughness of the plates and the finite temperature correction to the Casimir effect.

The first expression for the temperature Casimir force between two plates was obtained by Lifshitz,[11] and was then followed by some authors within the limits of quantum field theory in terms of the free energy density of vacuum.[12,13] To date, the Lifshitz formula in the case of the finite temperature still faces serious problems in theory,[14−16] and the temperature Casimir effect has not yet been measured. Fortunately, recent experiments on the lateral Casimir force open new opportunities for the test of the thermal Casimir effect,[17] if one chooses a suitable plate separation.

The originally predicted Casimir force between two flat plates is, for symmetry reasons, oriented normal to the surfaces of the plates. In fact, there is also a lateral Casimir force if both the plates are deformed, which was predicted in Refs. [18,19] and confirmed experimentally in Ref. [17]. The results in Ref. [20] show that the lateral Casimir force strongly depends on the ratio of the corrugation length λ and the plate separation H. Therefore one can choose the optimal parameters λ and H, and then test the thermal Casimir effect by measuring the lateral Casimir force.

In this Letter, based on the path-integral method,[18] the corrections to the Casimir energy due to the combined effect of the surface roughness and the finite temperature are calculated. For the specific case of sinusoidally corrugated plates, the lateral Casimir force at finite temperature is obtained, where the amplitude of the lateral Casimir force has a maximum at an optimal wavelength of $\lambda \approx 2H$ with the mean plate distance H.

We first calculate the Casimir energy between two parallel perfectly conducting plates at finite temperature by the path-integral quantization method. Similar to the treatment of waveguide geometries, the vacuum electromagnetic field in the vacuum between the plates can be decomposed into the TM and TE modes, and one can resort to a scalar field path-integral quantization.[21,23] Since the plates are assumed to be ideally conducting, the boundary conditions for TM and TE modes are of Dirichlet and Neumann types, respectively, i.e., $\phi_{\rm TM}|_S = 0$ and $\partial_n \phi_{\rm TE}|_S = 0$, where ∂_n denotes the normal to the plate surface S. In this study, we only discuss the Dirichlet boundary condition with $\phi|_S = 0$. After a Wick rotation to the imaginary time variable $\tau = it$, the scalar field ϕ can be quantized using the Euclidean action, and the corresponding partition function will be[20−24]

$$Z = \frac{1}{Z_0} \int_{\rm periodic} [D\phi] e^{\frac{-1}{\hbar} \int_0^\beta d\tau \int dx^3 \frac{1}{2}[(\partial_\tau \phi)^2 + (\nabla \phi)^2]} \cdot \prod_{\alpha=1}^{2} \prod_{\boldsymbol{y}} \delta[\phi(X_\alpha(\boldsymbol{y}))], \quad (1)$$

where k is Boltzmann's constant, the reciprocal of the temperature $\beta = (kT)^{-1}$ is the periodic length in the Euclidean time axis, and Z_0 is the partition function of the space without plates. 'Periodic' means that the functional integral is to be evaluated over field configuration satisfying $\phi(0, y_1, y_2, z) = \phi(\beta, y_1, y_2, z)$ with the 4D Euclidean space coordinates $X = (\tau, \boldsymbol{r}) \equiv (\tau, y_1, y_2, z)$. In Eq. (1), the Dirichlet boundary condition is enforced by the δ function $\prod_{\alpha=1}^{2} \prod_{\boldsymbol{y}} \delta(\phi(X_\alpha(\boldsymbol{y})))$, where we take $\alpha = 1, 2$ corresponding to the two plates. The surfaces of the two plates are parametrized by $\boldsymbol{y} = (y_1, y_2)$ with four-dimensional coordinates $X_1 = (\tau, y_1, y_2, 0)$ and $X_2 = (\tau, y_1, y_2, H)$. H is the separation distance between the two plates.

Using the integral representation of the δ functions,[25] the partition function of the system can

* Supported by the National Basic Research Programme of China under Grant No 2003CB716300, and the National Natural Science Foundation of China under Grant No 10505009.

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be rewritten as

$$Z=\frac{1}{Z_0}\int_{\text{periodic}}[D\phi]e^{\frac{-1}{\hbar}\int_0^\beta d\tau\int dx^3\frac{1}{2}[(\partial_\tau\phi)^2+(\nabla\phi)^2]}\cdot\prod_{\alpha=1}^{n=2}\int D[\varphi_\alpha(\boldsymbol{y})]e^{i\int d^2y_\alpha\phi_\alpha(\boldsymbol{y})\phi(X_\alpha(\boldsymbol{y}))},\tag{2}$$

with the auxiliary field $\varphi_\alpha(\boldsymbol{y})$ defined on the surface of the two plates. Since the scalar field ϕ is a function of τ defined on $\tau\in(0,\beta)$, we could use the Fourier series with

$$\phi(\tau,\boldsymbol{r})=\sum_{l=0}^{\infty}\sqrt{\beta}\phi_l(\boldsymbol{r})\sqrt{\frac{2}{\beta}}\cos\Big(\frac{2\pi l\tau}{\beta}\Big)+\sum_{l=1}^{\infty}\sqrt{\beta}\phi_l(\boldsymbol{r})\sqrt{\frac{2}{\beta}}\sin\Big(\frac{2\pi l\tau}{\beta}\Big).\tag{3}$$

Noting that the free energy, or Casimir energy H_{eff} is related to the partition function through the relation $Z=e^{-\beta H_{\text{eff}}}$, and substituting Eq. (3) into Eq. (2), we obtain

$$e^{-\beta H_{\text{eff}}}=\frac{1}{Z_0}\prod_{l=-\infty}^{\infty}Z_l,\tag{4}$$

with

$$\begin{aligned}Z_l=\int[D\phi_l(\boldsymbol{r})]\prod_{\alpha=1}^{n=2}[D\varphi_\alpha(\boldsymbol{y})]&\cdot\exp\Big[-\frac{\beta}{2}\int d^3\boldsymbol{r}d^3\boldsymbol{r}'\phi_l(\boldsymbol{r}')\\&\cdot\Big[\nabla\nabla'+\Big(\frac{2\pi l}{\beta}\Big)^2\Big]\delta^3(\boldsymbol{r}-\boldsymbol{r}')\phi(\boldsymbol{r})\\&+i\int d^2\boldsymbol{y}\varphi_\alpha(\boldsymbol{y})\phi(\boldsymbol{r}(\boldsymbol{y}))\Big].\end{aligned}\tag{5}$$

By introducing the Green function

$$G_l(\boldsymbol{r}-\boldsymbol{r}')=\int\frac{d^3\boldsymbol{p}}{(2\pi)^3}\cdot\exp[i\boldsymbol{p}\cdot(\boldsymbol{r}-\boldsymbol{r}')]\cdot\frac{1}{(\boldsymbol{p})^2+(2\pi l/\beta)^2},\tag{6}$$

satisfying $(\nabla\nabla'+(2\pi l/\beta)^2)G_l(\boldsymbol{r}-\boldsymbol{r}')=\delta^3(\boldsymbol{r}-\boldsymbol{r}')$, the integration over the field $\phi(\boldsymbol{r})$ in Eq. (5) can be performed using the Gaussian integral formula, leading to

$$\exp(-\beta H_{\text{eff}})=\prod_{l=-\infty}^{\infty}\int\prod_{\alpha=1}^{2}[D\varphi_\alpha(\boldsymbol{y})]e^{\frac{-1}{2\beta}(\varphi_\alpha,G_l(\boldsymbol{r})\varphi_\alpha)}.\tag{7}$$

Here we have introduced the following scalar product:

$$(\varphi_\alpha,G_l(\boldsymbol{r})\varphi_\alpha)=\sum_{\alpha=1}^{2}\sum_{\beta=1}^{2}\int d^2\boldsymbol{y}_\alpha d^2\boldsymbol{y}_\beta\varphi_\alpha(\boldsymbol{y})\cdot G_l(\boldsymbol{r}_\alpha(\boldsymbol{y})-\boldsymbol{r}_\beta(\boldsymbol{y}))\varphi_\beta(\boldsymbol{y}).\tag{8}$$

Finally, the effective Casimir energy between the two plates is obtained as

$$H_{\text{eff}}=\frac{1}{2\beta}\sum_l\ln\text{Det}M_l(\boldsymbol{y},\boldsymbol{y}'),\tag{9}$$

where the matrix M is a functional of $\boldsymbol{y}$ and $\boldsymbol{y}'$ and can be read from Eq. (8) as

$$M_l(\boldsymbol{y},\boldsymbol{y}')=\frac{1}{\beta}\begin{pmatrix}G_l(\boldsymbol{y}-\boldsymbol{y}',0)&G_l(\boldsymbol{y}-\boldsymbol{y}',H)\\G_l(\boldsymbol{y}-\boldsymbol{y}',H)&G_l(\boldsymbol{y}-\boldsymbol{y}',0)\end{pmatrix}.\tag{10}$$

The matrix $M_l(\boldsymbol{y},\boldsymbol{y}')$ describes the two flat plates and is only a function of the differences of the coordinates. It can be diagonalized by transforming to Fourier space,

$$M_l(\boldsymbol{p},\boldsymbol{p}')=\frac{1}{\beta}\begin{pmatrix}G_l(\boldsymbol{p},0)&G_l(\boldsymbol{p},H)\\G_l(\boldsymbol{p},H)&G_l(\boldsymbol{p},0)\end{pmatrix}\cdot(2\pi)^2\cdot\delta^2(\boldsymbol{p}+\boldsymbol{p}'),\tag{11}$$

with the Fourier-transformed Green functions

$$\begin{aligned}G_l(\boldsymbol{p},H)&=\int G_l(\boldsymbol{y},H)\exp(i\boldsymbol{p}\cdot\boldsymbol{y})d^2\boldsymbol{y}\\&=\frac{\exp\Big(-\sqrt{p^2+\Big(\frac{2\pi l}{\beta}\Big)^2}H\Big)}{2\sqrt{p^2+\Big(\frac{2\pi l}{\beta}\Big)^2}}.\end{aligned}\tag{12}$$

The determinant of $M_l(\boldsymbol{p},\boldsymbol{p}')$ can be easily computed, leading to

$$\begin{aligned}H_{\text{eff}}\equiv\sum_{l=-\infty}^{\infty}H_l&=\frac{kT}{2}A\sum_{l=-\infty}^{\infty}\Big(\int\frac{d^2\boldsymbol{p}}{(2\pi)^2}\ln\frac{G_l(\boldsymbol{p},0)}{\beta}\\&\quad+\int\frac{d^2\boldsymbol{p}}{(2\pi)^2}\ln\Big[1-\Big(\frac{G_l(\boldsymbol{p},H)}{G_l(\boldsymbol{p},0)}\Big)^2\Big]\Big)\\&=\frac{kT}{2}A\sum_{l=-\infty}^{\infty}\Big(\int\frac{d^2\boldsymbol{p}}{(2\pi)^2}\ln\frac{1}{2\beta\sqrt{(\boldsymbol{p})^2+\Big(\frac{2\pi l}{\beta}\Big)^2}}\\&\quad+\int\frac{d^2\boldsymbol{p}}{(2\pi)^2}\ln\Big[1-\exp\Big(-2\sqrt{p^2+\Big(\frac{2\pi l}{\beta}\Big)^2}H\Big)\Big]\Big)\end{aligned}\tag{13}$$

with A the two-dimensional surface area of the plate. Since the first term in Eq. (13) is independent in the separation H, it does not play an important role and is usually removed in the framework of dimensional regularization.

Considering the two polarization modes (TE and TM) and introducing the variable $\chi=2\sqrt{p^2+(2\pi l/\beta)^2}H$, the Casimir force per unit area between two flat plates will be

$$F=-\frac{2}{A}\frac{\partial H_{\text{eff}}}{\partial H}=-{\sum_{l=0}}'\frac{kT}{4\pi H^2}\int_{2H\xi_l}^{\infty}\chi^2\frac{1}{e^\chi-1}d\chi,\tag{14}$$

where the prime adds a multiple $1/2$ near the term $l=0$ and $\xi_l=2\pi l/\beta$ represents the Matsubara frequencies. The result of Eq. (14) is the same as the expression in Ref. [7].

Now we apply the path-integral approach to calculate the lateral Casimir force between two parallel plates covered by cosinoidally corrugations with a mean separation H. The positions of the surfaces for the two plates in the four-dimensional Euclidean space are $X_1 = (\tau, y_1, y_2, h_1)$ and $X_2 = (\tau, y_1, y_2, H + h_2)$. We consider the specific profiles with $h_1(y_1) = a\cos(2\pi y_1/\lambda)$ and $h_2(y_1) = a\cos(2\pi(y_1+b)/\lambda)$, which are shifted relative to each other by length b, the amplitude a and the wavelength λ (see Fig. 1).

The Casimir energy in this case is similar to Eqs. (9) and (10), only replacing the matrix M_l by M_l^h, which contains the correctional effect of the deformations due to $h_\alpha(\boldsymbol{y})$,

$$M_l^h(\boldsymbol{y},\boldsymbol{y}') = \frac{1}{\beta}\begin{pmatrix} \mathcal{M}_1\mathcal{M}_1' G_l(\boldsymbol{y}-\boldsymbol{y}', h_1(\boldsymbol{y}) - h_1(\boldsymbol{y}')) & \mathcal{M}_1\mathcal{M}_2' G_l(\boldsymbol{y}-\boldsymbol{y}', H + h_2(\boldsymbol{y}') - h_1(\boldsymbol{y})) \\ \mathcal{M}_1'\mathcal{M}_2 G_l(\boldsymbol{y}-\boldsymbol{y}', H + h_2(\boldsymbol{y}) - h_1(\boldsymbol{y}')) & \mathcal{M}_2\mathcal{M}_2' G_l(\boldsymbol{y}-\boldsymbol{y}', h_2(\boldsymbol{y}') - h_2(\boldsymbol{y}')), \end{pmatrix}$$
$$\mathcal{M}_1 = \sqrt[4]{1+(\nabla h_1(\boldsymbol{y}))^2}, \quad \mathcal{M}_1' = \sqrt[4]{1+(\nabla h_1(\boldsymbol{y}'))^2}, \quad \mathcal{M}_2 = \sqrt[4]{1+(\nabla h_2(\boldsymbol{y}))^2} \quad \mathcal{M}_2' = \sqrt[4]{1+(\nabla h_2(\boldsymbol{y}'))^2} \tag{15}$$

where the gradient $\nabla h_\alpha(\boldsymbol{y})$ is taken with respect to the plane (space–time) coordinates.[20–23] For the small deformations $h_{1,2} \ll H$, the above matrix can be perturbatively evaluated by expanding the Green function in terms of deformation field as

$$M_l^h(\boldsymbol{y},\boldsymbol{y}') = M_l(\boldsymbol{y},\boldsymbol{y}') + \delta M_l^h(\boldsymbol{y},\boldsymbol{y}'), \tag{16}$$

where $M_l(\boldsymbol{y},\boldsymbol{y}')$ is the matrix for two flat surfaces, which is defined in Eq. (10); and $\delta M_l^h(\boldsymbol{y},\boldsymbol{y}')$ is the correction due to deformations $h_\alpha(\boldsymbol{y})$.

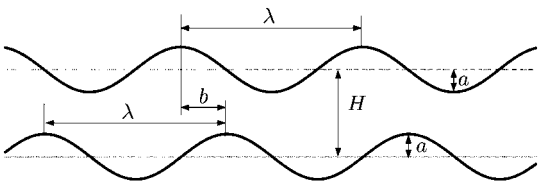


Fig. 1. Geometry used for calculating the lateral Casimir force between two corrugated plates with the amplitude a, wavelength λ and lateral shift b.

Following Refs. [15,17], we expand the partition function for the lth mode $\ln Z_l^h$ in a series $\ln Z_l + \ln Z_l|_1 + \ln Z_l|_2 + \cdots$, where the subscript indicates the corresponding order in $h_\alpha(\boldsymbol{y})$. The lowest-order result $\ln Z_l$ is defined in Eq. (5), which corresponds to two flat plates. The first order result $\ln Z_l|_1$ vanishes since we assume that the mean deformations are zero, $\int dy_1 h_\alpha(y_1) = 0$. The complete second-order contribution can be obtained by expanding $\ln\det(1 + M_l^{-1}\delta M_l^h)$ in powers of $h_\alpha(\boldsymbol{y})$ and keeping only up to second-order terms, which results in

$$\begin{aligned}\ln Z_l|_2 = &\int \frac{d^2\boldsymbol{p}}{(2\pi)^2}\cdot\frac{(p^2+\xi_l^2)\exp(-2\sqrt{p^2+\xi_l^2}H)}{[1-\exp(-2\sqrt{p^2+\xi_l^2}H)]^2}\\ &\cdot\int d^2\boldsymbol{x}[h_1^2(\boldsymbol{x})+h_2^2(\boldsymbol{x})]\\ &+\frac{1}{2}\int d^2\boldsymbol{x}\cdot d^2\boldsymbol{y}\cdot\Big[\Big(\int_{\xi_l}^\infty\frac{dp}{2\pi}p^2 J_0\\ &\cdot(|\boldsymbol{x}-\boldsymbol{y}|\sqrt{p^2-\xi_l^2}\Big)+\frac{K_1(l,t)}{H^3}\Big)^2\\ &+\Big(\frac{K_1(l,t)}{H^3}\Big)^2\Big]\cdot\Big[-\frac{1}{2}(h_1(\boldsymbol{x})-h_1(\boldsymbol{y}))^2\\ &-\frac{1}{2}(h_2(\boldsymbol{x})-h_2(\boldsymbol{y}))^2\Big]\\ &-\frac{1}{2}\int d^2\boldsymbol{x}\cdot d^2\boldsymbol{y}\cdot 2\frac{1}{H^6}K_2^2(l,t)\\ &\cdot(h_1(\boldsymbol{x})h_2(\boldsymbol{y})+h_1(\boldsymbol{y})h_2(\boldsymbol{x})),\end{aligned} \tag{17}$$

where the functions

$$\begin{aligned}K_1(l,t) = &\int_{(2\pi lH/\beta)}^\infty du\frac{u^2}{(2\pi)}\cdot\frac{1}{e^{2u}-1}\\ &\cdot J_0\big(t\sqrt{u^2-(2\pi lH/\beta)^2}\big),\\ K_2(l,t) = &\int_{(2\pi lH/\beta)}^\infty du\frac{u^2}{(2\pi)}\cdot\frac{e^u}{e^{2u}-1}\\ &\cdot J_0\big(t\sqrt{u^2-(2\pi lH/\beta)^2}\big),\end{aligned} \tag{18}$$

with the Bessel function J_0 and $t \equiv |\boldsymbol{x}-\boldsymbol{y}|/H$.

Substituting the specific profiles $h_\alpha(y_1)$ into Eq. (17), and inspecting the deformation dependent contributions to the Casimir energy, we obtain the energy E_c, which represents the corrugated-corrugated geometry and can yield the lateral Casimir force,

$$\begin{aligned}E_c = &\sum_l\cos\Big(\frac{2\pi b}{\lambda}\Big)a^2kT\frac{2\pi A}{H^4}\int_0^\infty ds\\ &\cdot\Big[\int_{(\frac{2\pi lH}{\beta})}^\infty du\frac{u^2}{(2\pi)}\cdot\frac{e^u}{e^{2u}-1}\\ &\cdot J_0\Big(s\sqrt{u^2-\Big(\frac{2\pi lH}{\beta}\Big)^2}\Big)\Big]^2\\ &\cdot sJ_0\Big(2\pi\frac{H}{\lambda}s\Big),\end{aligned} \tag{19}$$

which is dependent on the relative shift b.

Firstly, we consider the lateral Casimir force in $T = 0$. Under this condition, the summation can be replaced by integral and the energy that contributes the lateral Casimir force can be simplified to

$$\begin{aligned}E_c|_{T=0} = &Aa^2\cos\Big(\frac{2\pi b}{\lambda}\Big)\int ds\frac{\pi^2}{64H^5}\frac{\sinh^2 s}{\cosh^6 s}\\ &\times 2\frac{\sin\big(4sH/\lambda\big)}{4sH/\lambda}.\end{aligned}$$

Considering the two polarization modes, we obtain the lateral Casimir force per unit area at $T = 0$,

$$F_c\big|_{T=0} = -\frac{2}{A}\frac{\partial E_c|_{T=0}}{\partial b} = \frac{2\pi}{H^6}\Big(\frac{H}{\lambda}\Big)\sin\Big(\frac{2\pi b}{\lambda}\Big)j_0\Big(\frac{H}{\lambda}\Big), \tag{20}$$

with $j_0\Big(\dfrac{H}{\lambda}\Big) = \dfrac{\pi^2}{32}\int_0^\infty ds \dfrac{\sin(4sH/\lambda)}{4sH/\lambda}\dfrac{\sinh^2 s}{\cosh^6 s}$, which has the property $j_0(H/\lambda \to 0) \to 0$.

Then we consider the lateral Casimir force due to the finite temperature. Considering the two polarization modes, the lateral Casimir force per unit area at finite temperature can be expressed as

$$F_c = -\frac{2}{A}\frac{\partial E_c}{\partial b} = a^2\zeta\hbar c\frac{8\pi^2}{H^6}\Big(\frac{H}{\lambda}\Big)\sin\Big(\frac{2\pi b}{\lambda}\Big) \cdot \sum_l \int_0^\infty ds\Big[\int_{u_l}^\infty du\frac{u^2}{(2\pi)}\cdot\frac{e^u}{e^{2u}-1} \cdot J_0(s\sqrt{u^2-u_l^2})\Big]^2 s J_0\Big(2\pi\frac{H}{\lambda}s\Big) \quad (21)$$

with the dimensionless temperature $\zeta = kTH/\hbar c$. The lateral Casimir force at zero-temperature $\zeta = 0$ and finite-temperature $\zeta = 1.26$ are shown in Fig. 2.

The results of Eqs. (20) and (21) are thus exact to second order in a, and correctly take into account the many-body nature of the Casimir interaction, going beyond the commonly used pairwise summation for the van der Waals force.[20,26]

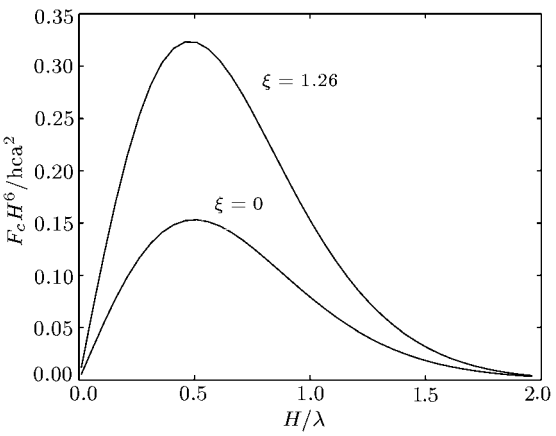


Fig. 2. Rescaled lateral Casimir force amplitude F_c at $\zeta = 0$ and $\zeta = 1.26$ (Taking $T = 300\,\mathrm{K}$ and $H = 10\,\mu\mathrm{m}$, one has $\zeta = 1.26$).

From Fig. 2, we know the optimum value of the parameter $H/\lambda \approx 0.5$, which is almost independent of temperature. At the room temperature $T = 300\,\mathrm{K}$, the amplitude of the lateral Casimir force per unit area is about $4\times10^{-8}\,\mathrm{N/m^2}$ with the mean separation $H = 10\,\mu\mathrm{m}$ and $a = 2\,\mu\mathrm{m}$. At present, we always use the torsion balance experiment to measure the weak forces. In the experiment, one can design the plates with the area $S = 10\,\mathrm{cm}\times10\,\mathrm{cm}$ and the beam length 10 cm. Then the torque will be about $4\times10^{-11}\,\mathrm{Nm^2}$. It could be measured since the present sensitivity of the torsion balance is about $10^{-15}\,\mathrm{Nm}$, which represents the thermal noise level for a general torsion balance.

From Eq. (21) we also know that for $\lambda/2 \gg H$ or $\lambda/2 \ll H$, the functions decay exponentially fast to zero so that the lateral force vanishes. It can be easy understood since both the two limit cases can go back to the flat geometry. Therefore, the optimum value of the parameter $\lambda/2$ must be the same order of magnitude with H. The numerical calculation of the optimum value $H/\lambda \approx 0.5$ is almost independent of temperature. In fact, the Casimir effect is one of geometric effect, which means that the main property of the force is relative to the geometry of the plates. The temperature usually affects the magnitude of the force directly, but changes the optimum value of H/λ. through the material thermal expansion and contraction, so it is not obvious to the optimum value of H/λ.

In addition, it should be noted that we have only calculated the Dirichlet boundary condition (TM mode) and have not considered the Neumann boundary condition (TE mode). We perform this study in the near future.

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